

EE111 ELECTRIC CIRCUITS

Midterm Exam, Spring 2026
回忆版

Question 1. For the circuit in Figure 1:

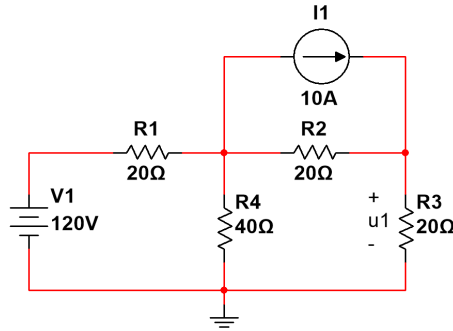


Figure 1.

- 1) Use Nodal analysis to find u_1 .
- 2) Find the power associated with the current source, and determine whether it is absorbing or supplying energy.

Question 2. For the circuit in Figure 2:

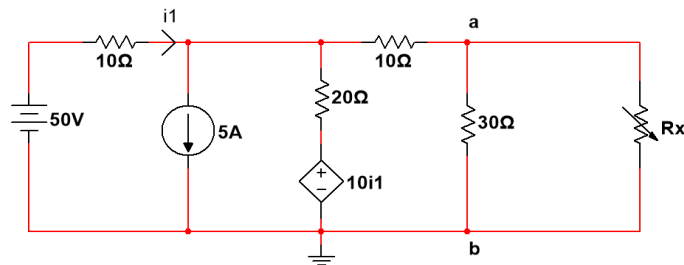


Figure 2.

- 1) Find Thevenin's equivalent circuit between terminal a, b (without R_x).
- 2) Determine the value of R_x that results in maximum power transfer to R_x , and calculate that maximum power.

Question 3. For the circuit in Figure 3:

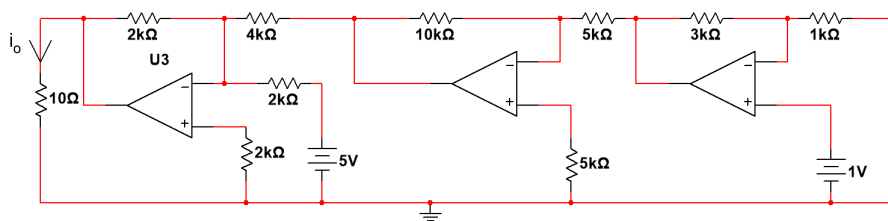


Figure 3.

All the operational amplifiers is ideal and operates in linear region. Find i_o in the circuit.

Question 4. For the circuit in Figure 4:

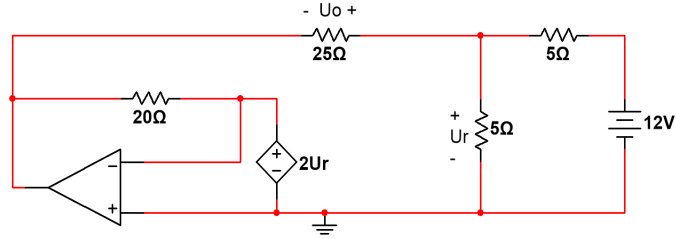


Figure 4

The operational amplifier is ideal and operates in linear region. Find U_o in the circuit.

Question 5. For the circuit in Figure 5, the expression $V_s(t)$ for the voltage source follows:

$$V_s(t) = \begin{cases} 6t + 15V, & 0 \leq t \leq 0.5s \\ 18V, & 0 > 0.5s \end{cases}$$

Assume the system reaches steady state before $t = 0$. The switch has been opened long enough before $t = 0s$, and it is closed at $t = 0s$.

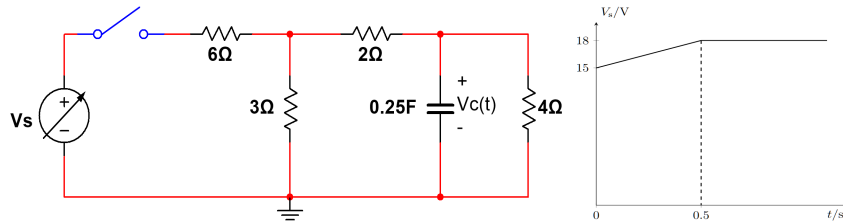


Figure 5

- 1) Find $V_c(t)$ for $t > 0$.
- 2) When $t \rightarrow +\infty$, find the energy stored in capacitor.

Question 6. For the circuit in Figure 6:

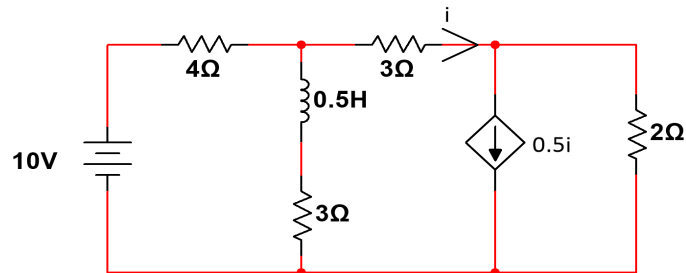


Figure 6

i_L is the current flow through the inductor. $i_L(0^-) = 3\text{ A}$, find $i_L(t)$ for $t > 0$.

Question 7. For the circuit in Figure 7:

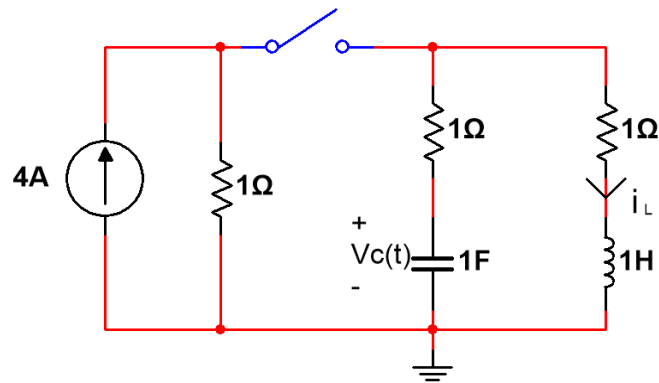


Figure 7

The switch is closed at $t = 0$, $V_c(0^-) = 3\text{ V}$, $i_L(0^-) = -1\text{ A}$. Find $V_c(t)$ for $t > 0$.